Quantifying the Economic Value of Specialty Pharmacy Care Through Pharmacist Intervention: A Cost Avoidance Study

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INTRODUCTION

- Specialty pharmacists monitor patients taking complex medications and help manage their therapy through targeted interventions focused on improving patient safety/tolerability, increasing treatment efficacy, and preventing medication waste.
- There is a potential to reduce health care costs by preventing suboptimal therapeutic responses and avoiding adverse outcomes that can be associated with specialty medications/conditions through these interventions.
- Accreditation standards for ACHC and URAC both include intervention documentation requirements. In addition, in recent years the NASP Clinical Outcomes Committee has focused on providing recommendations for clinical interventions, including recently publishing a white paper titled *Recommendations for Defining, Measuring, Reporting, and Associating Value with Clinical Interventions*.[1]
- These recommendations alongside prior studies have demonstrated how specialty pharmacies can approach an intervention framework and cost avoidance analysis to highlight the value their care model and clinical services provide.[2,3]

PURPOSE

This study aims to demonstrate the impact of the specialty pharmacy care model by calculating healthcare costs avoided as a result of pharmacist interventions in patients taking complex medications in a specialty pharmacy setting.

OUTCOMES

- **Primary Outcome**
  - Total avoided cost due to pharmacist interventions
- **Secondary Outcomes:**
  - Average cost avoidance by intervention type
  - Average cost avoidance by medication
  - Average cost avoidance by therapeutic category

METHODS

- Patients who were on specialty medications from January 2024 to July 2024 with a documented pharmacist intervention with an associated outcome of interest were included in this preliminary study.
- Two investigators reviewed data from a clinical management software.
  - Assigned potential outcome(s) and if the cost was direct and/or indirect.
  - For indirect costs, each investigator assigned the probability (Rare = 0.01 to Very Likely = 0.5) that the intervention would result in avoided downstream change.
  - To ensure a conservative estimate, the estimated indirect cost was multiplied by the associated probability.
  - Interrater variability was calculated utilizing percent agreement among the investigators.
- Direct costs were medication prices of tangible products and indirect costs were assigned based off the estimated cost of the avoided downstream consequence(s) using the Agency for Healthcare Research and Quality (HCUP) data.
- Drug costs were estimated using calculations described in the table below:

Type of Drug	Calculation		
	Estimate	Lower	Upper
Brand	AWP-20%	AWP-22%	AWP-18%
Generic	AWP-80%	AWP-85%	AWP-75%

Intervention Example:

Patient on TNFi diagnosed with pneumonia and advised to hold treatment. → Upon RPh follow-up, identified patient still has surplus on hand. → RPh advises appropriate delay to next fill date to avoid additional surplus and risk for waste. → Intervention documented to indicate significant impact on waste reduction (flag for investigator review).

Calculation Example:

Cost Avoidance Type	Estimated Avoided Cost (Month Supply of TNFi)			
Direct Cost	AWP	Estimate	Lower	Upper
	$2,000	$1,600	$1,560	$1,640
Indirect Cost		Likelihood		
		$49	0.01 = $16	0.05 = $82

CONTACTS AND DISCLOSURES

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None of the authors of this presentation have any financial relationships to disclose in relation to this project.

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